Practical 6:-

Code:-

```c
// philosophers.c

#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5

sem_t chopstick[N];
pthread_t philosopher[N];

void* eat(void* arg)
{
    int id = *(int*)arg;

    printf("Philosopher %d is thinking\n", id);
    sleep(1);

    sem_wait(&chopstick[id]);
    sem_wait(&chopstick[(id + 1) % N]);

    printf("Philosopher %d is eating\n", id);
    sleep(2);

    sem_post(&chopstick[id]);
    sem_post(&chopstick[(id + 1) % N]);

    printf("Philosopher %d finished eating\n", id);

    return NULL;
}

int main()
{
    int i, id[N];

    for (i = 0; i < N; i++)
        sem_init(&chopstick[i], 0, 1);

    for (i = 0; i < N; i++) {
        id[i] = i;
        pthread_create(&philosopher[i], NULL, eat, &id[i]);
    }

    for (i = 0; i < N; i++)
        pthread_join(philosopher[i], NULL);

    return 0;
}
```

Output :-

```
ASUS@LAPTOP-5DB0ADS1 MSYS ~
$ gcc philosopher.c -o philosophers

ASUS@LAPTOP-5DB0ADS1 MSYS ~
$ ./philosophers
Philosopher 0 is thinking
Philosopher 1 is thinking
Philosopher 2 is thinking
Philosopher 3 is thinking
Philosopher 4 is thinking
Philosopher 4 is eating
Philosopher 3 is eating
Philosopher 4 finished eating
Philosopher 2 is eating
Philosopher 3 finished eating
Philosopher 1 is eating
Philosopher 2 finished eating
Philosopher 0 is eating
Philosopher 1 finished eating
Philosopher 0 finished eating

ASUS@LAPTOP-5DB0ADS1 MSYS ~
$ |
```